

Session-7

16/09/2026

i) * Definition:

LCM

18

12

36

24

54

36

$$\therefore \underline{\underline{LCM = 36}}$$

Least Common multiple

HCF

$$18 = 2 \times 9$$

$$12 = 2 \times 6$$

$$= \underline{2} \times 3 \times 3$$

$$= 2 \times \underline{2} \times 3$$

$$= \textcircled{6} \times 3$$

$$= \textcircled{6} \times 2$$

Common

$$\therefore \underline{\underline{HCF = 6}}$$

Highest Common factor

* HCF and LCM:

HCF $\rightarrow$ 6

2	12	18
3	6	9
2	2	3
3	1	3
	1	1

$2 \times 3 \times 2 \times 3$
 $= 36 \rightarrow$ LCM

ii) • Basic-Method:

HCF $\rightarrow$ Common....
 LCM $\rightarrow$ 2, 3, 4, 5, 6, ...

Eg- $\rightarrow$ HCF

1	10	12	15	20	30
2	10	12	15	20	30
2	5	6	15	10	15
3	5	3	15	5	15
5	5	1	5	5	5
	1	1	1	1	1

LCM = 60

2	144	108	216
3	72	54	18
3	24	18	6
2	8	6	2
2	4	3	1
2	2	3	1
3	1	3	1
	1	1	1

$$\text{HCF} = 2 \times 3 \times 3 \times 2$$

$$= 36$$

$$\text{LCM} = 36 \times 4 \times 3$$

$$= \underline{\underline{432}}$$

- HCF is smaller or equal to given number.
- LCM is greater than given number.

iii) Bhramasra:

• Imp LCM List:

1) 6 & 8 $\rightarrow$ 24

2) 6 & 9 $\rightarrow$ 18

3) 6 & 12 $\rightarrow$ 12

4) 12 & 8 $\rightarrow$ 24

5) 12 & 9 $\rightarrow$ 36

6) 12 & 10 $\rightarrow$ 60

7) 12 & 14 $\rightarrow$ 84

8) 12 & 15 $\rightarrow$ 60

- 9) $12 \times 16 \rightarrow 48$
- 10) $12 \times 18 \rightarrow 36$
- 11) $12 \times 20 \rightarrow 60$
- 12) $12 \times 24 \rightarrow 24$ 13) $12 \times 27 \rightarrow 108$
- 14) $16 \times 14 \rightarrow 112$
- 15) $16 \times 15 \rightarrow 240$
- 16) $16 \times 18 \rightarrow 144$
- 17) $16 \times 20 \rightarrow 80$
- 18) $16 \times 24 \rightarrow 48$
- 19) $18 \times 14 \rightarrow 126$
- 20) $18 \times 15 \rightarrow 90$
- 21) $18 \times 20 \rightarrow 180$
- 22) $18 \times 24 \rightarrow 72$
- 23) $18 \times 27 \rightarrow 54$
- 24) $18 \times 30 \rightarrow 90$
- 25) $24 \times 15 \rightarrow 120$
- 26) $24 \times 20 \rightarrow 120$
- 27) $24 \times 30 \rightarrow 120$
- 28) $24 \times 36 \rightarrow 72$
- 29) $36 \times 27 \rightarrow 108$
- 30) $36 \times 30 \rightarrow 180$
- 31) $36 \times 40 \rightarrow 360$
- 32) $36 \times 48 \rightarrow 144$
- 33) $36 \times 45 \rightarrow 180$
- 34) $40 \times 45 \rightarrow 360$
- 35) $2 \quad 4 \quad 6 \quad 8 \quad 10 \quad 12 \rightarrow 120$
- 36) $10 \quad 12 \quad 15 \quad 20 \quad 30 \rightarrow 60$
- 37) $15 \quad 20 \quad 24 \quad 30 \quad 40 \rightarrow 120$
- 38) $16 \quad 18 \quad 24 \quad 36 \quad 48 \rightarrow 144$

* HW HCF & LCM

1) 144 216 288

2	144	216	288
2	72	108	144
3	36	36	72
3	12	12	24
2	4	4	8
2	2	2	4
2	1	1	2
	1	1	1

$$\text{HCF} = 2 \times 2 \times 3 \times 3 \\ = 36$$

$$\text{LCM} = 36 \times 8 \\ = 288$$

2) 308 196 140

HCF →

2	144	216	288
2	72	108	144
2	36	54	72
3	18	27	36
3	6	9	12
2	2	3	4
2	1	3	2
3	1	3	1
	1	1	1

HCF →

2	308	196	140
2	154	98	70
7	77	49	35
5	11	7	5
7	11	7	1
11	11	1	1
	1	1	1

$$\text{LCM} \rightarrow 72 \times 12 = \underline{864}$$

$$\text{LCM} \rightarrow 28 \times 5 \times 7 \times 11$$

$$\rightarrow 240 \times 7 \times 11$$

$$\rightarrow 1680 \times 11$$

$$\rightarrow 18480$$

$$\rightarrow 10780$$

3) 36 24 48 12 18

4) 40 60 80 20

HCF → 6

36	24	48	12	18
2	18	24	6	9
2	9	12	3	4.5
2	3	6	1.5	2
3	3	2	0.5	2
	1	1	1	1

HCF → 20

40	60	80	20
2	30	40	10
2	15	20	5
3	5	10	1
	1	1	1

$$\text{LCM} \rightarrow 6 \times 24 \rightarrow 144$$

$$\text{LCM} \rightarrow 20 \times 12 \\ \rightarrow 240$$

$$5) \quad 42 \quad 28 \quad 84 \quad 21$$

$$c) \quad 15 \quad 20 \quad 30 \quad 10$$

$$\begin{array}{l|rrrr} \text{HCF} \rightarrow 7 & 42 & 28 & 84 & 21 \\ \hline 2 & 6 & 4 & 12 & 3 \\ 2 & 3 & 2 & 6 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ \hline & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{array}{l|rrrr} \text{HCF} \rightarrow 5 & 15 & 20 & 30 & 10 \\ \hline 2 & 3 & 4 & 6 & 2 \\ 2 & 3 & 2 & 3 & 1 \\ 3 & 3 & 1 & 3 & 1 \\ \hline & 1 & 1 & 1 & 1 \end{array}$$

$$\text{LCM} \rightarrow 7 \times 12$$

$$\rightarrow \underline{\underline{84}}$$

$$\text{LCM} \rightarrow 5 \times 12$$

$$\rightarrow \underline{\underline{60}}$$

$$7) \quad 16 \quad 18 \quad 36 \quad 24 \quad 12$$

$$8) \quad 4 \quad 6 \quad 8 \quad 10 \quad 12$$

$$\begin{array}{l|rrrrr} \text{HCF} \rightarrow 2 & 16 & 18 & 36 & 24 & 12 \\ \hline 2 & 8 & 9 & 18 & 12 & 6 \\ 2 & 4 & 9 & 9 & 6 & 3 \\ 2 & 2 & 9 & 9 & 3 & 3 \\ 3 & 1 & 9 & 9 & 3 & 3 \\ 3 & 1 & 3 & 3 & 1 & 1 \\ \hline & 1 & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{array}{l|rrrrr} \text{HCF} \rightarrow 2 & 4 & 6 & 8 & 10 & 12 \\ \hline 2 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 1 & 3 & 1 & 5 & 3 \\ 5 & 1 & 1 & 1 & 5 & 1 \\ \hline & 1 & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{array}{l} \text{LCM} \rightarrow 48 \times 3 \\ \rightarrow 144 \end{array}$$

$$\begin{array}{l} \text{LCM} \rightarrow 8 \times 15 \\ \rightarrow 120 \end{array}$$

$$9) \quad 12 \quad 15 \quad 10 \quad 30 \quad 20$$

$$10) \quad 72 \quad 168 \quad 96 \quad 216$$

$$\begin{array}{l|rrrrr} \text{HCF} \rightarrow 1 & 12 & 15 & 10 & 30 & 20 \\ \hline 2 & 6 & 15 & 5 & 15 & 10 \\ 2 & 3 & 15 & 5 & 15 & 5 \\ 3 & 1 & 5 & 5 & 5 & 5 \\ 5 & 1 & 1 & 1 & 1 & 1 \end{array}$$

$$\text{LCM} \rightarrow \underline{\underline{60}}$$

$$\begin{array}{l|rrrr} \text{HCF} \rightarrow 3 & 72 & 168 & 96 & 216 \\ \hline 24 & 2 & 7 & 4 & 9 \\ 4 & 12 & 28 & 16 & 36 \\ 2 & 3 & 7 & 4 & 9 \\ 2 & 3 & 7 & 2 & 9 \\ 3 & 3 & 7 & 1 & 9 \\ 3 & 1 & 7 & 1 & 3 \\ 7 & 1 & 7 & 1 & 1 \\ \hline & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{array}{l} \text{LCM} \rightarrow 24 \times 3 \times 84 \\ \rightarrow 72 \times 84 \rightarrow 6048 \end{array}$$

$$11) 224 \ 168 \ 336 \ 280$$

$$12) 21 \ 24 \ 42 \ 14 \ 56$$

$$\begin{array}{c|ccccc} \text{HCF} \rightarrow 4 & 224 & 168 & 336 & 280 \\ 28 \times 2 & & & & \\ = 56 & 7 & 42 & 84 & 70 \\ & 2 & 6 & 12 & 10 \\ & 2 & 3 & 6 & 5 \\ & 2 & 3 & 3 & 5 \\ & 3 & 3 & 3 & 5 \\ & 5 & 1 & 1 & 5 \\ & & 1 & 1 & 1 \end{array}$$

$$\begin{array}{c|ccccc} \text{HCF} \rightarrow 1 & 21 & 24 & 42 & 14 & 56 \\ & 2 & 12 & 21 & 7 & 28 \\ & 2 & 6 & 21 & 7 & 14 \\ & 2 & 3 & 21 & 7 & 7 \\ & 3 & 7 & 1 & 7 & 7 \\ & 7 & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{aligned} \text{LCM} &\rightarrow 8 \times 3 \times 7 \\ &\rightarrow 8 \times 21 \\ &\rightarrow 168 \end{aligned}$$

$$\begin{aligned} \text{LCM} &= 56 \times 12 \times 5 \\ &= 56 \times 60 \\ &= 3360 \end{aligned}$$

$$13) 27 \ 18 \ 12 \ 9 \ 36$$

$$14) 32 \ 24 \ 12 \ 16 \ 8$$

$$\begin{array}{c|ccccc} \text{HCF} \rightarrow 3 & 27 & 18 & 12 & 9 & 36 \\ & 2 & 9 & 6 & 3 & 12 \\ & 2 & 9 & 3 & 2 & 6 \\ & 3 & 9 & 3 & 1 & 3 \\ & 3 & 3 & 1 & 1 & 1 \\ & & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{array}{c|ccccc} \text{HCF} \rightarrow 4 & 32 & 24 & 12 & 16 & 8 \\ & 2 & 8 & 6 & 4 & 2 \\ & 2 & 4 & 3 & 3 & 2 \\ & 2 & 2 & 3 & 3 & 1 \\ & 3 & 1 & 3 & 3 & 1 \\ & & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{aligned} \text{LCM} &\rightarrow 36 \times 3 \\ &\rightarrow 108 \end{aligned}$$

$$\begin{aligned} \text{LCM} &\rightarrow 24 \times 4 \\ &\rightarrow 96 \end{aligned}$$

$$15) 36 \ 18 \ 45 \ 70 \ 72$$

$$16) 15 \ 24 \ 20 \ 12 \ 30$$

$$\begin{array}{c|ccccc} \text{HCF} \rightarrow 2 & 36 & 18 & 45 & 70 & 72 \\ & 2 & 18 & 9 & 35 & 36 \\ & 2 & 9 & 9 & 10 & 18 \\ & 2 & 9 & 9 & 5 & 9 \\ & 3 & 3 & 3 & 15 & 3 \\ & 3 & 1 & 1 & 5 & 1 \\ & 5 & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{array}{c|ccccc} \text{HCF} \rightarrow 1 & 15 & 24 & 20 & 12 & 30 \\ & 2 & 15 & 12 & 10 & 15 \\ & 2 & 15 & 6 & 5 & 15 \\ & 2 & 15 & 3 & 5 & 15 \\ & 3 & 5 & 1 & 5 & 1 \\ & 5 & 1 & 1 & 1 & 1 \end{array}$$

$$\text{LCM} \rightarrow 24 \times 5 = 120$$

* Fractions-1:• Bonus (extra):

eg-1) 4 8 12 8 24 3 2
 LCM $\rightarrow$ 24 $\leftarrow$

2) 8 8 12 16 24 4 3

I) Basic $\checkmark$

II) Bhramstra $\checkmark$

III) Table start of bigger no. 24×2 & 16×3

$\therefore$ LCM = 48

3) 16 12 15 20 30

I) Basic $\checkmark$

II) Bhramstra $\checkmark$

III) Table start of bigger no. 30×2 , 12×5 , 20×3

LCM $\rightarrow$ 60

4) 8 8 16 8 36 18 48 12
 36 48

I) Basic $\checkmark$

II) Bhramstra $\checkmark$

III) Table start of bigger no $\rightarrow$ if table is hard then use 4th method advanced basic method

2	36	48
18	24	
72		

$72 \times 2 \rightarrow$ 144 LCM

3	36	48
12	16	
48		

$3 \times 48 \rightarrow$ 144

4	36	48
9	12	
36		

$4 \times 36 \rightarrow$ 144

6	36	48
6	8	
24		

$4 \times 24 \rightarrow$ 144

12	36	48
3	4	
12		

$12 \times 12 \rightarrow$ 144

• fractions :

$$1) \frac{7}{12} + \frac{17}{12} - \frac{1}{12}$$

$$= \frac{7+17-1}{12}$$

$$= \frac{23}{12}$$

$$2) \frac{5}{6} + \frac{7}{8}$$

$$= \frac{20+21}{24}$$

$$= \frac{41}{24}$$

$$3) \frac{11}{12} + \frac{7}{10} + \frac{13}{15} + \frac{17}{20}$$

$$= \frac{55+42+52+51}{60}$$

$$= \frac{200}{60}$$

$$= \frac{10}{3}$$

$$4) \frac{11}{12} + \frac{5}{6} + \frac{7}{8} + \frac{19}{24}$$

$$= \frac{44+40+42+38}{48 \text{ or } 24}$$

$$= \frac{164}{48}$$

$$= \frac{41}{12}$$

$$5) \frac{17}{24} + \frac{5}{6} + \frac{13}{18} + \frac{9}{8} + \frac{8}{9}$$

$$= \frac{51+60+52+81+64}{72}$$

$$= \frac{228+80}{72}$$

$$= \frac{308}{72}$$

$$= \frac{77}{18}$$

$$6) \frac{1}{2} + \frac{3}{4} + \frac{5}{6} + \frac{7}{8} + \frac{11}{12}$$

$$= \frac{12+18+20+21+22}{24}$$

$$= \frac{93}{24}$$

$$= \frac{31}{8}$$